

# Emily L. Hunt – Curriculum Vitae

📞 On request    ✉️ [emily.hunt.physics@gmail.com](mailto:emily.hunt.physics@gmail.com)    🌐 [emily.space](https://emily.space)    🔄 [emilyhunt](https://emilyhunt.github.io)

## Education & Employment

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**2025-29, Postdoc, University of Vienna, Austria**

**2025-25, Postdoc, Max Planck Institute for Astronomy, Germany**

**2023-24, Postdoc, Heidelberg University, Germany**

**Ph.D. 2023, Heidelberg University, Germany**

Thesis: “Improving the census of open clusters in the Milky Way with data from Gaia”

Advisor: S. Reffert

**M.Phys. 2019, University of Bath, United Kingdom**

Thesis: “Inference of photometric galaxy redshifts with a mixture density network”

Advisor: S. Wuyts

## Selected Presentations

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|----------------------------------------------------------------------|------|
| <b>Colloquium</b> – University of Vienna, Austria                    | 2024 |
| <b>Talk</b> , .Astronomy 12 – Flatiron Institute, New York, NY, USA  | 2023 |
| <b>Colloquium</b> , Königstuhl Colloquium – MPA, Heidelberg, Germany | 2023 |
| <b>Invited talk</b> , EAS (SS34) – Valencia, Spain                   | 2022 |
| <b>Invited talk</b> , EAS (S32) – Leiden, Netherlands                | 2021 |

## Open-source software

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**The Astrosky Ecosystem** – founder & lead developer of [The Astrosky Ecosystem](#), which is building astronomy outreach and communication tools for Bluesky/AT Protocol applications that are used by over 10,000 people a day

**ocelot** – lead developer of an upcoming [star cluster analysis Python package](#)

**The Star Formation Newsletter** – built a new, arXiv API-linked website for the Star Formation Newsletter’s [2026 relaunch](#)

## Teaching

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|------------------------------------------------------------------------------|------|
| <b>Python Programming for Scientists</b> , University of Vienna              | 2026 |
| <b>ISM &amp; Star Formation*</b> , University of Vienna                      | 2025 |
| <b>Machine learning*</b> , MWGaia Dr. Schl., University of Coimbra, Portugal | 2024 |
| <b>Astronomy Lab Course</b> , Heidelberg University                          | 2021 |
| <b>Introduction to Astronomy I</b> , Heidelberg University                   | 2020 |

\* = as a primary lecturer

## Supervision

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|-------------------------------------------------|---------|
| MSc supervisor: Janine Jochum                   | 2026-27 |
| BSc supervisor: 2 total                         | 2026-   |
| PhD committee: Kayvon Sharifi (Atlanta, GA, US) | 2026-   |

## Awards

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|------------------------------------------------------------------------------------------|--------------|
| <b>Ernst Patzer Award</b> for an excellent publication ( <a href="#">press release</a> ) | €2000 – 2023 |
| <b>University of Bath IMI Undergraduate Research Internship</b>                          | £2000 – 2018 |

## Selected Outreach

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|--------------------------------------------------------------------------------|------|
| <b>Invited talk</b> – OUTer SPACE, Max Planck Institute for Astronomy          | 2023 |
| <b>Interviewed for article</b> – Space.com                                     | 2021 |
| <b>Interviewed for article</b> – Thrillist.com                                 | 2020 |
| <b>Radio interview</b> – Deutschlandfunk (public radio) & Neue Zürcher Zeitung | 2020 |

## Meeting organization & service

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|--------------------------------------------------------------|---------|
| <b>Technical Editor</b> for the Star Formation Newsletter    | 2026-   |
| <b>Co-Chair</b> at EAS 2025: Symposium S3 (Cork, Ireland)    | 2025    |
| <b>SOC</b> for Roman Galactic Plane Survey Workshop (online) | 2025    |
| <b>SOC</b> for .Astronomy 13 (Madrid, Spain)                 | 2024    |
| <b>SOC</b> for .Astronomy 12 (New York, NY, USA)             | 2023    |
| <b>Reviewer</b> for A&A, ApJ, AJ, MNRAS                      | ongoing |

## Relevant expertise

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### Programming languages

**Python:** expert  
**JavaScript:** advanced  
**C/C++:** intermediate  
**Java:** basic

### Tools and scripting languages

**Git/GitHub:** expert  
**LaTeX:** expert  
**ADQL/SQL:** expert  
**HTML/CSS:** intermediate

## Languages

**English:** native speaker

**German:** intermediate

## Publications

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ADS search 

### First author

6. **Emily L. Hunt**, Vincent S. Carpenter, Kyle W. Cook *et al.* (2026). “The Astrosky Ecosystem: An independent online platform for science communication and social networking”. [ADASS XXXV \(Conference proceedings\)](#)
5. **Emily L. Hunt**, Tristan Cantat-Gaudin, Friedrich Anders *et al.* (2026). “The selection function of the Gaia DR3 open cluster census”. [A&A, 706, A341](#)
4. **Emily L. Hunt**, Tristan Cantat-Gaudin, Friedrich Anders *et al.* (2025). “The completeness of the open cluster census towards the Galactic anticentre”. [A&A, 699, A273](#)
3. **Emily L. Hunt** and Sabine Reffert (2024). “Improving the open cluster census. III. Using cluster masses, radii, and dynamics to create a cleaned open cluster catalogue”. [A&A, 686, A42](#)  
(175 citations)
2. **Emily L. Hunt** and Sabine Reffert (2023). “Improving the open cluster census. II. An all-sky cluster catalogue with Gaia DR3”. [A&A, 673, A114](#)  
(336 citations)
1. **Emily L. Hunt** and Sabine Reffert (2021). “Improving the open cluster census. I. Comparison of clustering algorithms applied to Gaia DR2 data”. [A&A, 646, A104](#)  
(129 citations)

### Co-author

6. Kristen C. Dage, **Emily L. Hunt** *et al.* (2026). “The Secret Lives of Open Clusters: a Multiwavelength Examination of Three Open Clusters”. [PASA, accepted](#)
5. Alexis L. Quintana, **Emily L. Hunt**, and Hanna Parul (2025). “How many stars form in compact clusters in the local Milky Way?”. [A&A, 701, L2](#)
4. Richard I. Anderson and **Emily L. Hunt** (2025). “A birds-eye view of stellar evolution through populations of variable stars in Galactic open clusters”. [A&A, 700, L13](#)
3. Sebastian Ratzenböck, João Alves, **Emily L. Hunt** *et al.* (2025). “Toward the fabric of the Milky Way: I. The density of disk streams from a local  $250^3$  pc<sup>3</sup> volume”. [A&A, 694, A307](#)
2. Dane Spaeth, Sabine Reffert, **Emily L. Hunt** *et al.* (2024). “Non-radial oscillations mimicking a brown dwarf orbiting the cluster giant NGC 4349 No. 127”. [A&A, 689, A91](#)
1. Cameren Swiggum *et al.* (incl. **Emily L. Hunt**) (2024). “Most nearby young star clusters formed in three massive complexes”. [Nature, 661, 8019, p.49-53](#)  
(25 citations)